Human Compression: current audit protocol

Human Compression project

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1 Corpus and current scope

The active corpus consists of 106 accessible originals from a fixed historical catalogue of 161 titles. New-title intake and retrieval of the 55 archived unavailable originals are outside the current programme. An accessible original remains a mathematical target while its recovery is unfinished. A complete authored rendering can supply a usable source without a publisher PDF, provided its edition and representation are identified explicitly.

The current partition is

$$106 = 91 \text{ reviewed corrected recoveries} + 9 \text{ principal refutations} + 5 \text{ pending principal dispositions} + 1 \text{ editorial}.$$

L022 is the editorial: complete reading found no principal mathematical result to recover. It remains an accessible original, with no measurement factor or mathematical recovery credit. The other 55 historical originals remain archived. The register counts originals, not theorem nodes, contracts, editions or files. Of the 91 corrected recoveries, 53 have accepted manuscript measurements. N014, L024, L102, F011, F008, L039, L013, L011, L025, L122, L015, L096, L023, L031, L017, L018, N011, F002, L071, L038, F006, F001, N012, L044, L059, L019, L042, L120, L045, L052, N010, L041, L027, L076, L100, L033, L070 and L091 are scientifically accepted but unmeasured; none is included in a new factor. Their corrections adjudicate the declared principal contracts, without asserting equivalence with every literal source statement. The goal is complete reviewed source disposition of all 105 mathematical originals, including surviving claims after a refutation, then stop. A refuted principal statement is handled separately from a recovery. The nine refuted originals retain surviving-claim obligations; together with the five pending originals, fourteen originals still require completion work. The count of 100 principal dispositions does not assert complete recovery of 100 sources.

F003 has a separate reviewed refutation of Theorem 1.1 of the preprint arXiv:2010.04080v1, by Pablo Spiga. The complete five-page proof constructs a chiral polytope of rank four and type $\{31, 31, 25\}$ for Sz(32768). The new marking and its proof are distinguished from borrowed Suzuki structure and polytope reconstruction. Exact Python arithmetic and an independent GAP construction check the finite data. This is one handled original, with no ordinary recovery, compression factor, new open-conjecture or Lean credit. No claim about the journal text, later editions or historical priority is made.

L029 has a separate reviewed refutation of Theorem 1.1 in the published 2021 paper *On residual connectedness in chiral geometries*. The complete five-page proof gives a rank-four geometry with 112 objects and 4368 chambers. The specified group is $\mathrm{PSL}_2(13)$ of order 1092; its full type-preserving group has order 2184 and two alternating chamber orbits. An actual rank-two residue is two disconnected six-cycles. Every hypothesis of the literal right-coset construction is verified. The rank-three proposition and specified-group variant are not refuted. No ordinary recovery, measurement, new-question or Lean credit is added.

L110, L114 and L115 have three separately reviewed principal refutations, with complete proofs of three, four and four pages. L110's literal Construction 11 omits a rank-four subfield

pregeometry. L114's rank-three list omits a thick, flag-transitive, residually connected geometry; its separate rank-two uniqueness assertion fails by the L115 family. L115 omits scalar-subspace covers: at $q = 32768$ the exhibited family alone has 41 equivalence classes under automorphisms of the specified group, exceeding the printed total 14. No bare-incidence count or replacement complete classification is claimed. Retained classical inputs, earlier witness provenance and all source-specific qualifications remain explicit. No ordinary recovery, measurement, new-question or Lean credit is added.

L027 and L076 add two corrected principal recoveries, with complete proof collections of 34 and six pages. Their exact source conventions, retained classifications and unreproduced historical census boundaries remain explicit. The three $\mathrm{PSU}_3(17)$ pairs are positive examples, not a classification; the $q = 81$ source correction acquires an earlier theorem. Measurement remains deferred.

L036, L046, L063 and L064 add four separately scoped principal refutations. For L036 the lower bound $50669 > 1511$ refutes the fixed Ru total without claiming a corrected exact total or recovering the other eleven rows. L046 applies the later disposition convention to its unchanged published correction: $798370 > 796772$, with type $(20, 31)$ exactly 9118. All 116 other exact cells, the exact total, chirality and finer inversion obligations remain unpaid; the old pending/no-credit decisions remain immutable historical records. L063 and L064 share one scalar-plane omission, with separately pinned source clauses; the latter target is the proof of Proposition 22, not Lemma 21. Nine planes are not nine geometry classes. None of these refutations is an ordinary recovery or a claim that all valid source assertions have been recovered. There is no new measurement, question-answer or Lean credit.

F008 is an auxiliary original, not focal-group authored. Its general-weight and improved-error ancillary remarks remain unreproved, and Conjecture 1.1 remains unanswered.

L039 retains its full three-page companion and seven-page rank-four replacement, with nine named external-input classes. Its source-recorded higher-rank chiral question remains unanswered; the corrected proof is not a claim of literal theorem-set equivalence.

L013 retains its full five-page general theory and four-page finite census, with three required mathematical programs and seven data files. The complete solver execution is a finite exhaustion premise; saved positive rows alone do not prove absence. Corrected endpoints and the unanswered future direction remain explicit, without a degree-six census or general small-field nonexistence claim.

L011 retains both whole four-page papers, all three required finite programs and their complete receipts. Eight negative cases are exhausted without a rank cutoff; eleven orders are freshly checked from generator-only groups. The positive classification remains a named retained input, and the auxiliary commutation statement is qualified to distinct projectivities. No new factor is inferred.

L025 retains the complete seven-page application, five-page common paper and three-page odd-cycle companion. Its 41 source dispositions correct the literal graph identification, semilinear map and edge-counter defect while preserving all principal cycle and extremality results. The four-cage completeness and small even-degree census remain named external premises. Two required programs and six data files remain explicit; the source listings on pages 7 and 15 require matched software treatment before any factor. No new question or Lean claim follows.

L122 retains all thirteen native pages: two positive papers, the actual coset counterexample and an explicit current contract. All 36 dispositions preserve the exact source hypotheses. The unrestricted triality-existence lemma is refuted; its counterexample is nonfirm and not residually connected. The qualified positive construction remains valid. Two required programs and two data files are whole premises; historical source software is unknown. No new factor or question follows.

L015 retains all fourteen native pages: four of new generation theory, six of application and four of unchanged inherited coset theory. Its 45 dispositions correct unrestricted generation and

the resulting ambient-connectivity claims while recovering the principal geometry and hypermap existence conclusions. Classical structure remains a named retained input. No finite program is a proof premise; one whole optional validator and one driver are disclosed, and historical software costs remain unknown. No factor or question credit follows.

L096 retains its complete eleven-page shadow common paper, nine-page application and seven-page naming companion, together with the entire existing six-page Topic D common. Its corrected contract preserves all 73 source dispositions and 232 named occurrences. Source profiles settle 229; three historical J_1 ordinals remain unverified metadata while the full mathematical profile classification is given. Complete required programs, data, checkpoints and retained classification inputs remain visible. No factor follows.

L023 acquires the complete three-page current contract and five whole scientific components of four, three, one, five and three pages. All 70 source dispositions and twelve finite rows are explicit. The odd-PSL family is uniformly completed; false unrestricted assertions remain refuted. Its broad final question is answered by separately attributed later L015 work, without new question credit. Required code and retained classifications remain visible. No comparison topic or factor is created.

L031 acquires seven whole native papers, totalling 23 pages, with the earlier four-page L011 obstruction retained byte-for-byte. The exact binary existence boundary, binary intersection counterexample and plus-type intermediate group correct the corresponding source clauses. All larger-field ranks, both Witt types and the complete low-dimensional spectrum are recovered. All 57 source dispositions retain explicit classical proof costs; no finite computation is a premise. Source questions remain recorded without new answer or status claim. No measurement, comparison topic or Lean declaration is added.

L017 retains all four complete papers (16 pages), 53 source dispositions and seven finite examples. The $q = 4$ graph distinction, $q = 7$ absent 84-vertex example and corrected full automorphism order remain explicit. Two exact programs and literal data are finite premises; classical inputs and source software costs remain declared. No factor, new question answer or Lean declaration is added.

L018 acquires both complete papers (8 pages), all 36 source dispositions, seven required programs and twelve whole data/receipt files. The seven-piece bound and both finite exclusion ranges are proved. Source arithmetic and perimeter qualifications remain explicit, as do the two unanswered general questions. No factor, new question answer or Lean declaration is added.

N011 acquires both complete papers (2+7 pages) and all 57 source dispositions. Both finite linear classifications and all twelve table rows are recovered on their intended domain $n \geq 2$, with intermediate literal corrections disclosed. External classification inputs remain retained; optional symbolic validation is not a finite proof premise. No factor, question or Lean claim is added.

F002 recovers the unique rank-four or rank-five polytope for even transitive degree-eleven groups: the self-dual eleven-cell. Both complete new papers (2+2 pages), the full acquired projective identification, and the required finite census and literal comparison data remain available. Five source graphs and ten orbit entries are checked. Auxiliary graph lemmas and the 52-tuple appendix are not separately recovered; all forty named statements have explicit dispositions. No factor, question answer or new Lean claim.

L071 recovers the principal mathematical classifications and search arguments of Leemans and Mixer, with two explicit source corrections. The chosen permutation action must realize all automorphisms by conjugation. H3 has eleven unrestricted classes up to duality, including three disconnected classes; the printed eight are the connected ones. All A11–A14 and Co3 counts and all 6,047 original sporadic/Tits markings are covered. Historical timings and the general alternating-rank question receive no new result claim. No factor or Lean credit is added.

L038 recovers every rank from 3 to $\lfloor (n-1)/2 \rfloor$ for all alternating groups of degree $n \geq 12$. The three complete papers contain 17 pages; finite end groups, signed permutation groups and

acquired rank contraction cover every congruence and endpoint. Three earlier maximal constructions remain explicit theorem inputs. Some auxiliary types differ from the source. All 3,200 finite subset-pair checks passed; the infinite arguments are written proofs. No measurement, new conjecture result or Lean claim is added.

F006 recovers the full D_n rank spectrum and all 42 cells in the connected Coxeter table. The immediate papers contain 21 pages, with the full acquired proofs retained. The original unrestricted rank claims require explicit corrections for I2(m), H3 and E7. Complete exceptional censuses contain 32,603 ordered markings, checked through 14,050,496 subset pairs. The written search proofs and terminal executions establish completeness. No factor, new conjecture result or complete Lean proof is claimed.

F001 recovers the complete rank spectrum of the projective symplectic group in dimension four over every finite field. A separate counterexample corrects an auxiliary orthogonal subgroup lemma; the principal theorem holds. N012 recovers all twelve principal classifications and both tables in the 2025 linear group thesis, with an explicit correction to the centre structure in Corollary 4.2.1. Their complete proofs, retained inputs and distinct component reviews are available through their publication guides. Neither recovery adds a compression factor, an open-conjecture claim or a complete Lean proof.

L044 recovers the exact maximal rank of the natural alternating group in every degree $n \geq 3$. Complete finite bases and independent lower constructions join the primitive, imprimitive and intransitive omission arguments in a strict induction. The auxiliary independent-generator claim in Proposition 3.3 requires the explicit narrower correction. The fourteen immediate manuscripts are entry points to the full retained proof closure; their 37 pages are not a compression measurement. No new open-conjecture or Lean credit is claimed.

L059 recovers the faithful graph-action classification on 1950 inputs, with 4254 markings and all 39 printed cells, including the published cubic subdivisions. Its eight source qualifications, external catalogue premises and two unanswered classification questions remain explicit. The fifteen pages of immediate manuscripts lead to the complete proof and validation collection; no compression factor or new Lean result is claimed.

L019 recovers the stable rank-and-degree correspondence for $n \geq 2\kappa + 3$, with marked isomorphism and reversal, and the six constants $1, 1, 7, 9, 35, 48$. Actual contraction and complete rooted wing searches pay the inverse and finite boundary. Four auxiliary source corrections and the degree-six action qualification remain explicit. The nine immediate pages lead to the full acquired proof closure; no compression factor or new Lean result is claimed.

L042 recovers all eighteen explicit families of rank $n - k$ for $1 \leq k \leq 4$ and $n \geq 2k + 3$, with counts $1, 1, 7, 9$. The symmetric-group conclusion and intransitive maximal omissions are qualified in degree six by the ordinary simplex action. Its four-page application acquires the complete L019 proof and finite census; no measurement or new conjecture credit follows.

L120 recovers the fixed classification survey as an attributed synthesis. Twenty-five of its thirty-two assertion clauses have acquired or corrected routes; seven named published theorem inputs remain explicit. Of its 221 rank-count cells, 212 have project certificates and nine retain the independent L026 classification. All six tables, eighteen diagram families and eighteen question dispositions are mapped. The nine-page synthesis retains the complete acquired closure and adds no measurement or conjecture credit.

L045 recovers the positive simple-group theorem and its chiral-polyhedron consequence. Its seven-page synthesis acquires 51 complete component pages, with all finite certificates and distinct reviews. The 36 source dispositions separate the positive theorem from excluded-family negatives, the announced PSL3/PSU3 converse, historical map counts and the almost-simple question. Named classification inputs remain attributed. The 58-page reading volume adds no compression measurement.

L052 recovers the complete corrected general and finite hypertope contract. Its five-page synthesis acquires the complete eight-page general proof and five-page finite proof. The 62 source

dispositions retain 23 finite table entries, three examples, four corrections and four named classifications. Questions 9.1, 9.2 and chiral-table completeness remain unanswered. The ancillary A6 rotation illustration is corrected without a new general C-plus converse. The 18-page reading volume adds no compression measurement.

N010 gives a complete attributed disposition of the fixed historical survey: 16 contexts, 52 clauses, 34 numbered problems, 14 secondary records and 66 bibliography entries. The eleven-page synthesis and four complete components form a 22-page reading volume; older acquired proofs remain additional. Retained classifications and source-only totals 17 and 441 are not independently reproduced. The 137-quotient result concerns one family only. Unanswered questions retain their exact limits, with no claim of current problem status, new question credit or compression measurement.

L041 gives a complete eight-page decorated-point and tetrahedral-family proof, routing all twenty numbered source statements in twenty-eight dispositions. Coordinate, period and ancillary corrections remain explicit. Internal residue identification is used only after actual triple-flag transitivity in this family. Triangular Euclidean presentation and acquired chamber theory remain named inputs. Five finite fields are controls only, with no new measurement, question credit or formal declaration.

L100 gives a complete corrected recovery of the fixed 22-page Halle preprint on geometries for $\mathrm{PSL}_3(4)$. The seven-page synthesis and ten component papers give 36 immediate proof pages, with editable native inputs and a portable finite certificate. All 55 source clauses are routed: 29 geometry classes, their complete structural data, and all construction and truncation joins. Eight retained premises and six ancillary corrections remain explicit, including the displayed butterfly and oval qualifications. The 25-page journal edition is not collated. Local primitivity, rather than local two-transitivity, defines the search boundary. No new measurement, question answer, principal refutation or Lean declaration is added.

L033 gives a complete corrected recovery of the fixed sixteen-page revised manuscript on locally spherical regular hypertopes. Its nine-page synthesis and twelve component papers provide 50 immediate proof pages. All 109 source clauses have explicit dispositions, including 22 named finite positives, four excluded printed rows and nine bounded minimum interfaces. Eight retained classical and computational input classes and seven correction groups remain explicit. Diagram-only spherical residues and universal residues have separate minimum statements; two universal conclusions are exclusion bounds only. No new measurement, question answer, principal refutation or Lean declaration is added.

L070 gives a complete corrected recovery of the fixed twelve-page article on two atlases of abstract chiral polytopes. Its sixteen complete manuscripts form a 41-page volume including a four-page synthesis. All 46 source clauses have explicit dispositions. The first atlas has 56 examples, with arbitrary cores retained; Table 1 names and the stated vertex figures are certified. The second atlas has 79 named groups and 88 printed occurrences, with corrected totals 9,210 regular and 16,974 oriented chiral classes. Source orders, Aut-group names and historical conventions are joined explicitly. The larger printed domain omits $\mathrm{Sp}_4(4)$; no polytope census for that omitted group or historical program execution is inferred. Classical and software inputs remain explicit. No new measurement, question answer, principal refutation or Lean declaration is added.

L091 gives a complete corrected disposition of the fixed six-page Higman–Sims article and its acquired supplement. Its eleven-page synthesis and twenty-six accepted proof chapters form a 78-page volume, including one divider. The proof join covers all 86 source diagrams, 398 local structure occurrences, 312 parabolics and the full automorphism and correlation groups of all 86 geometries. The original rank counts survive; source corrections, blank fields and our own transcription repairs are separate. Five prescribed doublings are proved, while three incompatible source prescriptions retain explicit qualifications. The eight comparison rows preserve their equivalence conventions. J3 global coverage and absence of higher ranks remain

attributed to L091 reference [9], L097; that external original receives no recovery credit. No measurement, new conjecture answer, principal refutation or Lean declaration is added.

The accepted overview v19 reports nineteen complete-topic factors and fifty-three conditional application factors. Its graphs show the whole common theory and all applications. The exact current records are the corpus register, the overview pointer and the manuscript catalogue. The six-page overview gives the values; this protocol adds no new measurement or scientific admission.

2 The mathematical object being recovered

Before comparing lengths, each original receives an edition-bound source contract. Its principal results, parameter ranges, exceptional cases, geometric identifications and computational obligations are stated explicitly. A complete application must prove the relevant hypotheses of the common theory and supply all local arguments needed for those outputs. Retained classical theorems, classification tables and catalogue constructions remain named premises.

A false source assertion is recorded as a correction, not as a theorem proved with its original wording. Unrecovered ancillary statements, illustrations, classifications and open questions remain explicit. Recovery of the principal contract does not assert reproduction of every assertion in the source. Consequently, a full-source length comparison is a comparison for the declared principal outputs, not a claim that every source passage has an equivalent replacement passage. The source still contributes its complete counted manuscript, including ancillary exposition and references.

Scientific acceptance, input acceptance and numerical acceptance are separate. An author packet, distinct review and later admission retain their exact bytes and roles. Earlier candidate labels in a frozen packet are interpreted through its subsequent admission. Existing proofs can be reorganized or reused, but moving a proof between files does not itself establish a new mathematical result.

3 Whole topics and conditional applications

For a topic T, write $L_1, \dots, L_k$ for the complete source manuscripts, U_T for its entire common theory, and $A_1, \dots, A_k$ for its complete applications. An application includes every required written companion. Under the topic's fixed manuscript counting convention c_T, the whole factor is

$$R_T = \frac{\sum_{i=1}^{k} c_T(L_i)}{c_T(U_T) + \sum_{i=1}^{k} c_T(A_i)}.$$

All common papers and additions actually used in the topic belong to U_T. The whole common theory is charged once within that topic. Separate topics pay their own common theories in full, even when their arguments overlap. There is no cross-topic pooled factor or unrecorded discount for previously written work.

The individual comparison is

$$r_{T,i} \mid \mathcal{K}_T = \frac{c_T(L_i)}{c_T(A_i)}.$$

Here $\mathcal{K}_T$ includes the *whole* common theory, the declared required programs and data, and any retained mathematical premises. Their acquisition costs and unknown costs are reported separately. Any additional sibling-application premise must be included explicitly in the acquisition boundary; it cannot be supplied for free. The accepted current A edition, for example, pays all 41 common pages across three complete papers and all ten applications in its whole-topic comparison, rather than allocating only selected common paragraphs to each one.

A factor greater than one records shortening in the stated coordinate; a factor below one records expansion. A conditional saving may coexist with a whole-topic expansion. Such results are retained without changing the metric or dropping applications to obtain a preferred value. These factors are not measures of historical effort, mathematical originality or minimum proof complexity.

4 Text representations and sensitivity

The input protocol is fixed and reviewed before factors are evaluated. It binds the exact source edition, every complete replacement PDF, document boundaries, software masks, font evidence, normalization and units. Titles, abstracts, body text, captions and references remain included. Exact running furniture can be removed only in the declared selected-text coordinates; an untrimmed coordinate preserves it. Some accepted topics use that untrimmed coordinate as primary. The overview retains each topic's accepted choice rather than silently replacing them with a single new convention.

The usual sensitivity family combines raw untrimmed, raw selected and calibrated selected text with NFC and NFKC normalization and five units: whitespace-separated words, alphanumeric runs, non-whitespace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. The exact definitions and any additional representation sensitivities are those of each frozen input protocol. A display-rounding choice changes no exact fraction. A near-unit factor may reverse under another accepted unit; this is reported rather than suppressed.

Calibration identifies the actual PDF font dictionary and encoding, the text event and the visible glyph. Font names alone need not identify an encoding. Repairs, including visible glyphs extracted as whitespace, require occurrence and outline evidence. A chosen proxy is declared as a proxy. Raw extraction remains available for sensitivity; calibration is not a transcription of all visual mathematical content.

Complete authored renderings, including the accepted N001 and L108 sources, are distinguished from unavailable original PDFs. Website navigation or duplicated reference widgets are not article text. Images, diagrams, layout and unextracted labels have unknown cost; their absence from text extraction is not a zero-cost estimate. In particular, the L035 source contains scientific image content not represented by its extracted text. All current factors are subject to their documented representation limits.

5 Symmetric software treatment and proof artifacts

The current scalar is manuscript-only. Concrete executable program and implementation-listing costs are excluded symmetrically from source and replacement. This rule applies to required computations as well as optional validation. Written mathematical algorithms and proof prose remain counted on both sides. A software role alone cannot justify counting a source listing while omitting the corresponding replacement program.

For L008, L028 and N006, independently reviewed masks remove the exact embedded implementation blocks, retaining surrounding mathematical statements and exposition. Their restored manuscript comparisons use those accepted boundaries. The listing audit and matched input packet preserve the locators and sensitivity evidence. Earlier one-sided software-inclusive values remain historical observations and are not current comparisons.

Required computations remain proof premises. Each topic inventories every whole required program, certificate, mathematical data file and decoder, with separate roles for optional validation, execution drivers and partial formal artifacts. Reused identical content may be counted once within an actual topic dependency closure, with every occurrence and role disclosed. No

file is pruned merely because a shorter word certificate or a previously run search is available. Retained external representations and catalogues remain explicit trust and acquisition boundaries.

A printed program appendix and its raw files are alternative representations, not two additive copies of one program. A printed appendix need not contain a required data file; the data still belongs in the complete inventory. Historical source software and full external proof costs are generally unknown. Accordingly, there is no current software-inclusive factor. No missing cost is interpreted as zero, and incompatible units such as pages and bytes are never added.

6 Proof dependencies and formal evidence

A logical graph records actual prerequisite-to-use relations with source and proof locators. Mathematical premises, definitions, retained theorems, computations, validation, attribution and operational dependencies have distinct roles. A citation's presence does not by itself prove logical dependence. A graph accepted for an earlier edition is not automatically a complete graph for an enlarged topic.

Where passage costs have been accepted, each counted passage has one owner. For an output t, let $D(t)$ be its prerequisite closure and $P(v)$ the passage owners used by node v. The known passage cost is

$$W_{\text{known}}(t) = \sum_{p \in \bigcup_{v \in D(t)} P(v),\ c(p) \text{ known}} c(p).$$

The union prevents repeated charging of shared passages. Unknown costs remain listed separately; context outside that closure is still paid in the whole-manuscript comparison. Parent and child views must preserve this total. Raw node counts are not compression factors.

For an explicit acquisition order and a previously acquired library K_{i-1}, a sequential account can report the new closure cost $c(D_i \setminus K_{i-1})$. It requires actual accepted dependencies and costs; it does not establish a sawtooth pattern in advance or alter separate topics' whole-common charges.

Lean evidence is limited to the actual compiled declarations and their stated hypotheses, dependencies and axiom reports. A theorem assuming an unproved recognition or geometric hypothesis does not formalize that hypothesis. Successful compilation and declaration counts do not establish complete source formalization. Finite certificates likewise require an explicit argument from the checked computation to the claimed result. The current numerical comparison adds no complete-Lean or conjecture-answer claim.

7 Reproduction and historical records

Current TeX/PDFs, source maps and admissions are routed through the public catalogue. Original source assets are separately supplied where redistribution is unavailable. Portable input and arithmetic checkers record their exact versions, hashes and output boundaries. The canonical builder uses each manuscript's declared compiler and full TeX inputs. Repository integrity checks, formal compilations and expensive finite replays are different operations; evidence states which actually ran.

Historical editions remain immutable. The preceding README and preceding protocol retain the old chronology and comparisons. The intake record, archived notes and earlier overview admissions remain evidence for their stated editions. A documentation update supersedes reader-facing status and policy statements without rewriting those records. Separately reviewed question answers retain their own exact scope; an absent historical proof has no measured length and does not produce an infinite compression factor.